

REVA Academy for Corporate Excellence

SRN	AI09								
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Name: Hari Krishnan

Program: MSc in Artificial Intelligence

Course: Computer Vision

Date: 27th September 2025

Time: 02:00pm to 5:00pm

Answers:

Section: A – UNIT I

Q2.

- Using a sample image of your choice (nature/portrait/product), generate and plot the histogram of pixel intensities in grayscale. Interpret what the histogram tells about brightness and contrast distribution.
- In manufacturing quality checks, filters are often used to detect defects. Write Python code to apply Gaussian blur, median filter, and sharpening filters on a product image (e.g., a circuit board). Compare the results and explain which filter best highlights fine defects and why.
- Develop a case study where edge detection (Sobel, Laplacian, Canny) is applied to an industry problem (e.g., detecting cracks in tiles or boundaries in satellite images). Implement code, visualize outputs, and critically evaluate which method is best suited.

Ans: Added in **AI09_Module8_End_Exam_Computer_Vision_HariKrishnan(1).py** file

Section: B – UNIT II

Q3.

- Use the Intel Image Classification dataset.
(<https://www.kaggle.com/datasets/puneet6060/intel-image-classification>)
(≈25k images, 6 categories like buildings, forest, street). Train a simple CNN classifier (3–4 convolution layers). Submit code, accuracy plots, and test results. In case the training is taking time, then take a small sample of images.
- Compare your CNN with a pretrained ResNet18 (fine-tuned). Report accuracy, confusion matrix, and discuss why transfer learning works better on smaller datasets.
- Assume this model is deployed in a smart city monitoring system (classifying scenes from CCTV feeds). What risks exist if the model misclassifies (e.g., forest → building)? Suggest mitigation strategies.

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Ans: Added in **AI09_Module8_End_Exam_Computer_Vision_HariKrishnan(2).pynb** file

Section: C – UNIT III

Q6.

a. Using the Oxford-IIIT Pet Dataset (<https://www.kaggle.com/datasets/tanlikesmath/the-oxfordiiit-pet-dataset>), apply semantic segmentation with a pretrained DeepLabV3 model from TorchVision. Visualize segmentation masks for at least 5 images.

b. Perform instance segmentation using a pretrained Mask R-CNN model on the same images. Compare semantic vs instance masks (e.g., when multiple cats/dogs appear).

c. Imagine deploying this system in a veterinary clinic to assist in analyzing pet scans/photos. Propose a workflow diagram, explain practical benefits, and highlight limitations of segmentation models in medical-like settings.

Ans: Added in **AI09_Module8_End_Exam_Computer_Vision_HariKrishnan(2).pynb** file

Section: D – UNIT IV

Q7.

a. Download any image from the web (e.g., a natural scene, an object). Use a pretrained Vision Transformer (ViT) from Hugging Face to classify the image. Show top 5 predicted classes with probabilities.

b. Using the same image, apply CLIP (Contrastive Language–Image Pretraining) with two different text prompts (e.g., “a photo of a dog” vs. “a photo of a cat”). Report similarity scores and analyze how CLIP bridges vision + language.

c. Imagine building a visual search engine for an e-commerce site (customers search with text, system retrieves matching product images). Propose a pipeline using ViT + CLIP. Draw workflow diagram and critically evaluate potential challenges (e.g., bias, misclassification).

Ans: Added in **AI09_Module8_End_Exam_Computer_Vision_HariKrishnan(1).pynb** file

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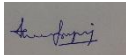
Time: 02:00pm to 5:00pm

STUDENT UNDERTAKING

I, **[Hari Krishnan]**, bearing **Student ID/Registration Number: [AI09]**, hereby declare that:

- All answers provided in this answer script are **original** and a result of my own understanding.
- I have not engaged in any form of **copying, plagiarism, impersonation, or malpractice**.
- I understand that **any academic dishonesty** will lead to disciplinary action as per university regulations.

Signature: _____



Date: _____27-Sept-2025_____

*****End*****